End Semester Examination

Classical Mechanics
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Duration: 180 minutes.
Total points: 80.

Please give arguments where necessary. If it is unclear from your answer why a particular step is being taken, full credit will not be awarded. Grades will be awarded not only based on what final answer you get, but also on the intermediate steps.

1. Consider Atwood's machine with a massive pulley. On the two sides of the pulley, there are two masses of mass m and $2m$ respectively. The pulley itself is a solid disc of mass m and radius r. The string connecting the masses and passing over the pulley is itself massless and does not slip over the pulley. See Figure 1. Find the acceleration of the masses.
   10 points

2. We begin with a stick of uniform density and length l. Systematically, one removes the middle third of the stick. Of the remaining two pieces on the two sides, one again removes the middle third (an amount of length $l/9$ from each). This process continues ad infinitum. Let the final object have a finite mass m (if you are worried about infinite density, there is no need to be). Find the moment of inertia of the final object about an axis perpendicular to the stick through its original center. (In case you are curious, this is the Cantor set, the prototypical one dimensional (to begin with) fractal object).
   10 points

3. A particle of mass m is thrown up vertically with an initial speed V_0, reaches a maximum height and falls back to the ground. In another instance, the same particle is dropped vertically downward from the same maximum height. In both instances, the particle will undergo a Coriolis' deflection. Compare the (a) direction and (b) magnitude of the Coriolis' deflection in the two instances when the particle hits the ground in both cases.
   10 points

4. A heavy point mass M is constrained to move on a massless hoop of radius R which stands vertically above the ground and rotates about the vertical

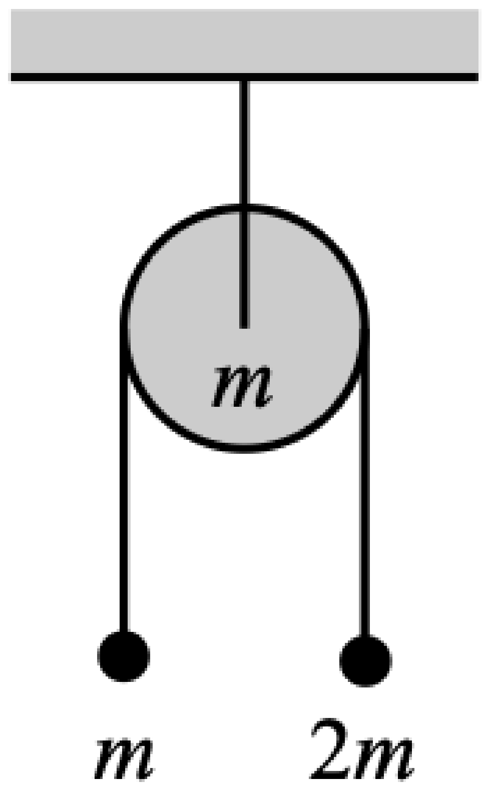

Figure 1: Massive Atwood's Machine

axis with a constant angular frequency ω. The only external force acting on the mass-hoop system is gravity.

(a) Derive the Lagrangian equations of motions.

(b) What are the conserved quantities?

(c) Show that there is a ω_0 such that if $\omega > \omega_0$, then there is one point other than the bottom where the mass can remain motionless, while for $\omega < \omega_0$, the only such point is the bottom of the hoop. Find ω_0.

5 + 3 + (5 + 2) = 15 points

5. Consider two masses m_1 and m_2. m_1 is constrained to move on the $z = 0$ plane in a circle of radius a centered at $x = y = 0$. m_2 is constrained to

move on the $z = c$ plane in a circle of radius b centered at $x = y = 0$. The two masses are connected by a light massless spring of spring constant k.

(a) Write down the Lagrangian of the system.

(b) Solve the problem using the method of Lagrange multipliers and give a physical interpretation of each multiplier.

5 + (7 + 3) = 15 points

6. Consider a stick of mass m and length l pivoted at the top. The stick is so arranged so as to always make an angle θ with the downward vertical. See Figure 2. The stick carries out rotational motion about the pivot. Find the angular velocity ω of this motion.
10 points

7. Suppose a disc rotates horizontally in a counterclockwise direction with an angular velocity of magnitude ω. A person, standing at a radius r from the origin of the disc, starts walking radially inwards, with a constant speed (relative to the disc) v.

   (a) Find the Coriolis force (both magnitude and direction) on the person while analyzing the problem in the frame co-rotating with the disc.

   (b) In the inertial (lab) frame, of course, there is no Coriolis force. Describe how the various physical forces on the person act to satisfy the equation $\frac{d\vec{L}}{dt} = \vec{\tau}_{\text{ext}}$.

   4 + 6 = 10 points

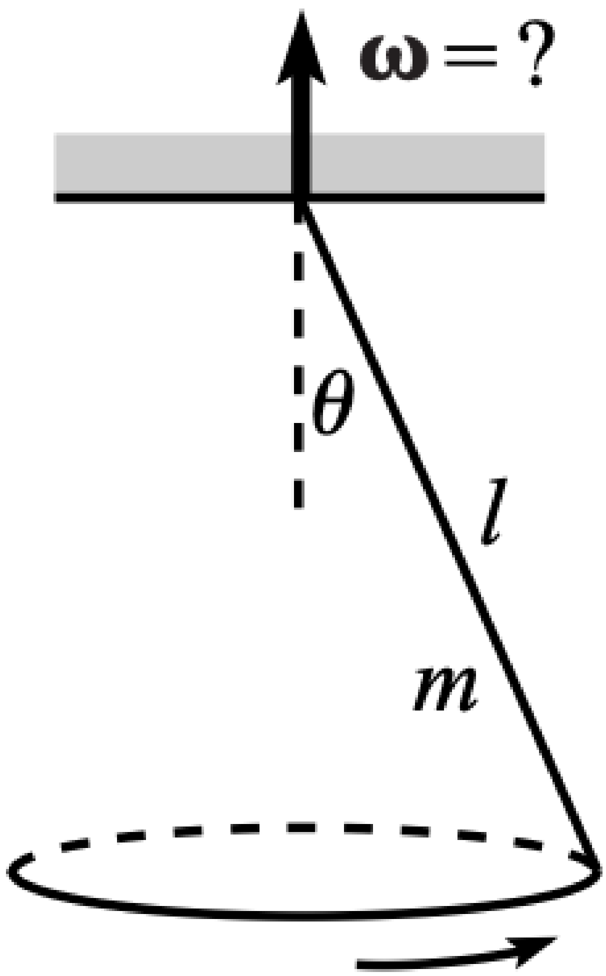

Figure 2: Massive stick rotating about a pivot